

A First-Principles ARK Treatment of the Classical Three-Body Problem Without Hilbert Space Formalism

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Abstract

The classical three-body problem is a deterministic system derived directly from least action in Newtonian gravity, and is therefore not intrinsically a Hilbert-space problem. Practical difficulties arise not from a lack of state-space language, but from the point-mass singularity and resulting stiffness during close encounters. This note presents a first-principles formulation consistent with the Aether Reality Kernel (ARK) framework in which gravitational interaction is represented by a scalar geometry field θ^c coupled to matter density ρ , and singular behavior is regularized by a constitutive stiffness response k^c that bounds curvature (and thus acceleration) in high-gradient regimes. Resonance and phase-locking behavior can be expressed natively using the ARK scalars (θ^c , τ^c , k^c , ω , dS_t) without importing Hilbert-space machinery.

1. Introduction

The three-body problem in celestial mechanics concerns the coupled motion of three masses under mutual gravitation. Its equations are well-posed and derivable from least action; the practical challenge is the absence of a general closed-form solution and the presence of strong sensitivity to initial conditions, especially during close approaches. In some discussions, additional mathematical formalisms (for example, Hilbert-space language) are introduced as if they were required to define or resolve the system. For the classical three-body problem, they are not: the system is classical, and the key technical issue is how one treats singular behavior implied by the point-mass idealization.

2. First-Principles Baseline: Least Action

For three point masses m_i at positions $r_i(t)$, a standard Newtonian Lagrangian is:

$$L = \sum_{i=1..3} (1/2) m_i || \dot{r}_i ||^2 + \sum_{i<j} G m_i m_j / || r_i - r_j ||$$

Stationary action $S = \int L dt$ yields the Euler-Lagrange equations:

$$m_i \ddot{r}_i = \sum_{j \neq i} G m_i m_j (r_j - r_i) / || r_j - r_i ||^3$$

This is the classical three-body problem: deterministic, continuous, and defined on ordinary configuration space (positions and velocities). No Hilbert space is required to state or evolve it.

2.1 The Real Source of the “Divide by Zero” Complaint

The apparent singularity arises from the $1/||r_i - r_j||$ term as the separation approaches zero. Point masses are an idealization; real bodies have finite size and internal response. A first-principles resolution therefore targets the constitutive behavior of the underlying medium/geometry under high curvature, rather than adding an extra entity (for example, a “void body”) or changing vocabulary.

3. ARK Geometry-First Formulation

ARK encodes interaction through a scalar geometry field $\theta^c(x,t)$ whose gradients produce acceleration. In the weak-field limit one may anchor θ^c to a Poisson-type source relation:

$$\nabla^2 \theta^c(x,t) = \lambda \rho(x,t)$$

Particle acceleration is defined from the local geometry gradient:

$$\ddot{r}_i(t) = -C \nabla \theta^c(r_i(t), t)$$

Here λ and C are calibration constants determined by the chosen normalization of θ^c and the Newtonian limit mapping. Up to calibration, this mirrors the familiar relation $a = -\nabla\Phi$ but expressed in ARK scalar terms.

4. Singularities Resolved by Constitutive Stiffness k^c

To remove blow-ups at close approach without patchwork, ARK introduces a constitutive response: the substrat has finite stiffness k^c , so curvature cannot diverge arbitrarily. A minimal nonlinear field law that captures stiffness-limited curvature is:

$$\nabla \cdot (\mu(||\nabla \theta^c|| ; k^c) \nabla \theta^c) = \lambda \rho$$

$\mu(\cdot)$ is a constitutive mobility of the geometry that decreases at high gradient so the flux of $\nabla \theta^c$ saturates. This produces Newton-like behavior in low-gradient regimes and bounded curvature in high-gradient regimes.

4.1 Example Saturation Law

One simple family of saturation laws is:

$$\mu(s; k^c) = 1 / (1 + (s / s_0(k^c))^n), \quad s = ||\nabla \theta^c||, \quad n \geq 1$$

For $s \ll s_0$, $\mu \approx 1$ and the equation reduces to the Poisson form (recovering Newtonian behavior). For $s \gg s_0$, μ rolls off and further steepening is suppressed.

4.2 Consequence: Bounded Acceleration

Since $\ddot{r} = -C \nabla \theta^c$, bounding $||\nabla \theta^c||$ bounds $||\ddot{r}||$. Close encounters therefore do not produce infinite acceleration; the instability is removed by a physical constitutive limit (finite stiffness) rather than an arbitrary softening epsilon. This is the core first-principles distinction: patchwork softening modifies a denominator; ARK stiffness specifies that infinite curvature is not physically admissible.

5. Resonance and Phase-Locking in Native ARK Scalars

Multi-body systems often exhibit resonance capture, phase-lock plateaus, and metastable configurations. In ARK these behaviors can be expressed natively using the coupled roles of:

- ω : phase/update anchor (supports stable cadence and phase-locked attractors)
- τ^c : tension memory (hysteresis/inertia enabling capture plateaus)
- dS_t : entropy delta (dissipative selection; determines persistence vs decay)
- θ^c : geometry slope field (drives interaction via gradients)
- k^c : stiffness (limits curvature; controls close-encounter thresholds)

In conservative idealizations one may set $dS_t \approx 0$. In real systems, small positive dissipation (tides, radiation, internal friction) selects attractors and can lock resonant configurations. What is sometimes described externally as “tuning-fork” locking is naturally represented by ω (phase anchoring) coupled with τ^c (memory) and dS_t (selection).

6. Practical Simulation Protocol (Toy Demonstration)

A clean demonstration compares standard Newtonian point-mass evolution to ARK-regularized evolution under identical initial conditions. Near close approach, explicit Newtonian integrators typically require extremely small adaptive timesteps or ad hoc softening to avoid numerical blow-up.

ARK-regularized evolution proceeds by (i) constructing $p(x,t)$ (for example, compact kernels around particles), (ii) solving the nonlinear field equation for θ^c , and (iii) updating particle motion from $-C\nabla\theta^c$. The reported outputs should include bounded accelerations during close approach, stability without ad hoc epsilon-softening, and resonance capture/escape behavior as a function of ω , τ^c , and dS_t .

7. Why Hilbert Space Is Not Required Here

Hilbert space is the natural setting for quantum state vectors and probability amplitudes. The classical three-body problem is defined on configuration space and derived from least action. Introducing Hilbert space does not supply missing first principles; it at most supplies an alternate representation. The actual technical issues are (i) sensitivity/chaos and (ii) singular behavior from point-mass idealization. ARK addresses (ii) at the constitutive level via k^c and expresses (i) as geometry-driven attractor sensitivity modulated by ω , τ^c , and dS_t .

8. Conclusion

We presented a first-principles ARK treatment of the classical three-body problem that (1) begins from least action, (2) recasts interaction through a scalar geometry field θ^c , (3) removes singular blow-ups through stiffness-limited curvature via k^c , and (4) captures resonance/locking behavior using native ARK scalars (ω , τ^c , dS_t). This avoids the conceptual overhead of Hilbert-space framing for a classically defined system while providing an explicit physical mechanism for regularization rather than an ad hoc patch.